

Abubakr Elmallah

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EDUCATION

University of California, Irvine

Irvine, CA

Bachelor of Science – Computer Science (3.90 GPA)

Expected June 2027

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Discrete Mathematics

WORK EXPERIENCE

iOS App Developer

March 2025 – Present

UCI Student Center | studentcenter.uci.edu

Irvine, CA

- Own and ship three production App Store apps ([UCI Now](#), [ZOTFinder](#), [UCI Esports](#)) for UC Irvine Student Center & Event Services, totaling **100K+** installs and **3.2M+** lifetime sessions.
- Cut load times **40%** by rebuilding UCI Now and ZOTFinder from **UIKit** into **SwiftUI** with **async/await** networking and a full redesign, extending both apps to **iPad**, **Mac**, and **Vision Pro**.
- Reduced ZOTFinder crashes **85%** (10.4K to under 1.5K/month) by refactoring the legacy codebase.
- Integrated 20+ **CocoaPods** libraries (**Alamofire**, **SwiftlyJSON**, **Google Maps**, **Firebase**) in the UIKit apps.
- Launched UCI Esports 1.0 as **sole iOS developer** by integrating **ggLeap REST APIs** for live PC reservations, **Google Calendar** event feeds, **UCINetID** sign-in with guest mode, and backend profile sync.

Lead Software Developer

September 2024 – Present

ICS Student Council (ICSSC) | studentcouncil.ics.uci.edu

Irvine, CA

- Lead a **12-person team** building **PeterPlate**, ICSSC's open-source campus dining platform, **live in production** on web and iOS, serving UCI students menu, allergen, nutrition, dish rating, and event data.
- Built a type-safe frontend with **Next.js (React)**, **TypeScript**, **Zustand**, **Tailwind CSS**, and **shadcn/ui**.
- Shipped the backend by pairing **tRPC** with **Drizzle ORM** on a 13-table **PostgreSQL** schema, deployed to **AWS Lambda** and **RDS** via **GitHub Actions** CI/CD, with **Docker** + **Testcontainers** integration tests.

iOS App Development Teacher

February 2024 – February 2026

STEM Heroes Academy | stemirvine.com

Irvine, CA

- Launched the academy's first **iOS App Development** course as its **youngest instructor**, teaching **Swift**, **Swift Playgrounds**, and **SwiftUI** to 30+ students (ages 8–14) who each shipped a working app over two years.

PROGRAMMING PROJECTS

[Aurebesh Translator App Suite](#) | iOS [Top 10 on App Store Entertainment Chart](#) [GitHub](#) | 2023

- Built and published my first iOS app at 17 in **SwiftUI**, growing it into free ([Aurebesh Translator](#)) and premium ([Datapad](#)) apps earning **\$1K+** from **25K+** installs and **60K+** sessions at **99.9% crash-free**.
- Enabled instant offline translation by engineering a modular engine rendering **14 galactic scripts**.
- Decoded photographed glyphs offline with an on-device **camera scanner** template-matching 14 bundled fonts.
- Shipped 6 **WidgetKit** widgets, **Metal** shader effects, **StoreKit** in-app purchases, and a watchOS companion app.

[Al-Islam App Suite](#) | iOS [Congressional & Swift Student Challenge Winner](#) [GitHub](#) | 2023

- Built **three full-featured SwiftUI apps** ([Al-Islam](#), [Al-Quran](#), [Al-Adhan](#)) for iPhone, iPad, Mac, and Apple Watch with **WidgetKit** widgets, serving **15K+** users across **55K+** sessions at **99.8% crash-free**.
- Cut load times **50%** across all three apps through **JSON parsing**, **SwiftSoup** scraping, and offline caching.

[Star Wars Discord Bots](#) | Python, C++ 2024 – 2025

- Grew [Sabacc Droid](#) (Python, [discord.py](#)) and [Aurebesh Droid](#) (C++20, D++, CMake) to **500+ Discord servers** by engineering an event-driven, 8-player multiplayer engine spanning 4 Sabacc card-game variants.
- Sustained **99.9% uptime** by deploying to **Heroku** with automated **GitHub Actions** builds.

LANGUAGES & TECHNOLOGIES

Languages: Swift, Python, C++ (C++20), TypeScript, JavaScript, SQL, Bash

Frameworks & Libraries: SwiftUI, UIKit, WidgetKit, StoreKit, WatchConnectivity, Next.js, React, Zustand, Tailwind CSS, shadcn/ui, tRPC, Drizzle ORM, Alamofire, Firebase, Google Maps SDK, discord.py, D++

Tools: Xcode, Git, GitHub Actions, Docker, AWS, PostgreSQL, Heroku, Vercel, Turborepo, CocoaPods, CMake